

El Noh

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EDUCATION

- 02/2026 **Ph.D.**, Department of Atmospheric Science, Kongju National University, South Korea
- *Thesis title*: Dynamical changes in mid-latitude circulation and associated increase in heatwave.
- 08/2020 **M.S.**, Department of Atmospheric Science, Kongju National University, South Korea
- *Thesis title*: Dynamical mechanism involved in the development of the western Pacific pattern.
- 08/2018 **B.S.**, Department of Atmospheric Science, Kyungpook National University, South Korea

RESEARCH EXPERIENCES

- 03/2026
– present **Postdoctoral Researcher**, Kongju National University, South Korea (Supervisor: Prof. Joowan Kim)
• Research topics: The role of Rossby waves in summer extreme weather; Dynamical mechanism of the variabilities in atmospheric energetics.
- 08/2018
– 02/2026 **Research Assistant**, Kongju National University, South Korea (Advisor: Prof. Joowan Kim)
• Research topics: The role of Rossby waves in summer extreme weather.
- 04/2023
–04/2024 **Research Associate**, Division of Atmospheric Sciences, Korea Polar Research Institute (Supervisor: Dr. Sang-Yoon Jun)
• Research topics: Numerical simulation of quasi-stationary Rossby waves in the North Pacific; Predicting marine heatwaves in East Asia.

ACADEMIC EXPERIENCES

- 2023 **Teaching Assistant** at Kongju National University
• *Fundamentals of Programming in 2023 fall semester*

HONORS & AWARDS

- 2019 Best Poster Presentation, Korean Meteorological Society Fall Conference

PUBLICATIONS (Peer-reviewed)

1. **Noh, E.**, Kim, J., Jun, S. Y., Pak, G., Kim, J. H., & Kim, H. G. (2023). Atmospheric pathway of marine heatwaves over the Northwestern Pacific. *Scientific Reports*, 13(1), 22821.
2. **Noh, E.**, Kim, J., Jun, S. Y., Cha, D. H., Park, M. S., Kim, J. H., & Kim, H. G. (2021). The role of the Pacific-Japan pattern in extreme heatwaves over Korea and Japan. *Geophysical*

Research Letters, 48(18), e2021GL093990.

3. H. M. Lee, **E. Noh**, J. H. Kim, J. Kim, & S. Y. Jun. (2024). Contribution Assessment of Northern Hemispheric Atmospheric Circulations to Korean Mid-Summer Surface Warming by the Atmospheric Nudging Experiment. *Asia-Pacific Journal of Atmospheric Sciences*, 60(2), 119-130.
4. J.-H. Kim, S.-J. Kim, D. Youn, M.-K. Kim*, J.-H. Kim, J. Kim, & **E. Noh** (2021). Impact of North Atlantic-East Asian teleconnections on extremely high January PM10 cases in Korea. *Environmental Pollution*, 290, 118051.
5. [In Korean] J.-H. Kim, **E. Noh**, & M.-K. Kim* (2021). The Relationship between the Arctic Oscillation and Heatwaves on the Korean Peninsula. *The Korean Journal of Quaternary Research*, 33(1-2), 25-35.

MANUSCRIPT IN PREPARATION

6. [Under Review] **E. Noh**, et al. (2026). European Heatwave Aggravated by Recent Summer Arctic Changes. *Nature communications*.
7. [Under Review] **E. Noh**, S. Y. Jun, S. W. Yeh, Y. Kosaka, J. Kim. (2026). A Process-Based Evaluation Metric for the Pacific–Japan Pattern (PJEM) and its Influence on Asian Summer Extreme.
8. [In preparation] **E. Noh**, S. S. Woo, Y. Kosaka, J. Kim. (2026). Precipitation enhancing processes by the Pacific-Japan pattern in East Asian summer monsoon.

INTERNATIONAL CONFERENCES

Oral Presentations

1. **El Noh**, Joowan Kim (2026) European Heatwave Exacerbated by Recent Arctic Changes, *European Geosciences Union (EGU) General Assembly*.
2. **El Noh**, Joowan Kim (2025) Emerging Changes in Boreal Summer Rossby Wave, *Busan IAMAS-IACS-IAPSO (BACO) Joint Assembly*.
3. **El Noh**, Joowan Kim (2024) The Role of the Meridional Rossby wave for Extreme Heatwaves Over East Asia, *European Geosciences Union (EGU) General Assembly*.

Poster Presentations

1. **El Noh**, Joowan Kim (2025) Summer Arctic warming and potential implication for European Heatwaves, *American Geophysical Union (AGU) Fall Meeting*.
2. **El Noh**, Joowan Kim (2025) Revisiting the Lorenz Energy Cycle: Impacts of Global Warming on Atmospheric Energy Cycle, *European Geosciences Union (EGU)*.
3. **El Noh**, Joowan Kim, Sang-Yoon Jun, Gyundo Pak, Joo-Hong Kim, Hyeong-Gyu Kim (2023) Atmospheric pathway of marine heatwaves over the Northwestern Pacific, *American Geophysical Union (AGU) Fall Meeting*.
4. **El Noh**, Joowan Kim, Sang-Yoon Jun, Gyundo Pak, Joo-Hong Kim. (2023) East Asian Heatwaves driven by Meridional Rossby waves, *28th International Union of Geodesy and*

Geophysics (IUGG) General Assembly.

5. **El Noh**, Joowan Kim, Sang-Yoon Jun, Dong-Hyun Cha, Myung-Sook Park, Joo-Hong Kim, Hyeong-Gyu Kim. (2022) The Role of the Pacific-Japan Pattern in Extreme Heatwaves Over East Asia, *23rd Conference of Atmospheric and Oceanic Fluid Dynamics (AOFD)*.